CR-4342297: KD Bus VPN Transition & SSL Certificate Deployment

CR Number	4342297	Request Date	2026-01-22
Requested By	Olli Ojala	Risk Level	LOW
Environment	Meraki Dashboard - KD BUS | Azure East & South East	Rollback	< 5 min
Change Window	TBD	Downtime	< 5 min during route switch

1. Summary

Purpose: Establish IPsec VPN connectivity between Meraki MX (ATE-TEST pilot site) and Azure FortiGate firewalls. Deploy SSL certificates and DNS records for FortiGate VPN endpoints and FortiClient EMS.

Scope: Pilot test only (ATE-TEST network). Production rollout via separate CR.

2. Infrastructure Details

Component	Azure East	Azure South East
FortiGate Hostname	VPN1.KeolisDowner.com.au	VPN2.KeolisDowner.com.au
FortiGate Public IP	68.218.113.249	4.200.139.87
FortiGate Inside IP	10.200.1.68	10.201.1.68
Azure Subnet	10.200.10.0/24	10.201.10.0/24
Route Table	KD-Prod-RouteTable	KD-Mel-RouteTable

EMS Server	Value
Hostname	EMS1.KeolisDowner.com.au
Public IP	20.11.121.242

Pilot Site	Value
Network	ATE-TEST
Subnet	10.160.10.0/24

3. SSL Certificate

Attribute	Value
Type	Wildcard SSL/TLS Certificate
Common Name	*.keolisdowner.com.au
Standard	X.509 v3, SHA-256, RSA 2048-bit minimum
EKU	TLS Web Server Authentication (OID: 1.3.6.1.5.5.7.3.1)

Deploy To	FortiGate East, FortiGate South East, FortiClient EMS

4. DNS Records Required

Hostname	Type	IP Address	Purpose
VPN1.KeolisDowner.com.au	A	68.218.113.249	FortiGate East VPN
VPN2.KeolisDowner.com.au	A	4.200.139.87	FortiGate South East VPN
EMS1.KeolisDowner.com.au	A	20.11.121.242	FortiClient EMS

5. Changes to be Made

System	Action
DNS	Create 3 A records (VPN1, VPN2, EMS1)
Certificate	Obtain *.keolisdowner.com.au wildcard cert from CA
FortiGate East	Import cert, configure IPsec Phase1/Phase2 for ATE-TEST
FortiGate SEast	Import cert, configure IPsec Phase1/Phase2 for ATE-TEST
FortiClient EMS	Import cert (PFX), restart service, update FG connectors
Meraki	Add fortigate-test tag to ATE-TEST, create 2 IPsec peers
Azure East	Add route 10.160.10.0/24 → 10.200.1.68 to KD-Prod-RouteTable
Azure SEast	Add route 10.160.10.0/24 → 10.201.1.68 to KD-Mel-RouteTable

6. IPsec VPN Parameters

Parameter	Value
IKE Version	IKEv2
Encryption	AES-256
Authentication	SHA256
DH Group	14 (2048-bit)
PFS	Enabled, Group 14
Phase 1 Lifetime	1800 seconds
Phase 2 Lifetime	3600 seconds
PSK	Shared securely outside CR (>32 chars)

7. Test Plan

- Ping from ATE-TEST (10.160.10.x) to Azure East (10.200.10.10) - expect reply
- Ping from ATE-TEST to Azure SEast (10.201.10.61) - expect reply
- RDP to Azure VMs to confirm no asymmetric routing
- Verify VPN tunnel stats incrementing on FortiGates

- Verify HTTPS to VPN1, VPN2, EMS1 hostnames shows valid certificate
- Confirm other KD BUS networks unaffected (no FortiGate tunnels visible)

8. Rollback Procedure

If tests fail: Execute in order (estimated < 5 minutes)

1. Remove route from KD-Prod-RouteTable (Azure East)
2. Remove route from KD-Mel-RouteTable (Azure SEast)
3. Disable/remove ATE-TEST VPN on both FortiGates
4. Delete IPsec peers in Meraki Dashboard
5. Verify ATE-TEST connectivity via existing SonicWall tunnels

9. Prerequisites

- Wildcard certificate *.keolisdowner.com.au obtained
- DNS A records created and propagated
- FortiGate firewalls forwarding to Azure backend
- NSG/Azure Firewall permits FortiGate inside IPs
- Meraki Dashboard access with Org write permissions
- PSK shared securely (>32 characters)

Full Runbook: CR-2026-001-Fortigate-IPsec-Test-Combined.html
Production Rollout: Separate CR required after successful pilot